

CHIRANTHAN SHADAKSHARASWAMY

in LinkedIn · GitHub · ☎ (812) 553-1460 · ✉ chiranthan.s19@gmail.com · 🌱 Skillfolio

SUMMARY

Software Development Engineer on AWS Transit Gateway (core Hyperplane team), the distributed cloud router that connects customers' VPCs, VPNs, and on-premises networks. End-to-end owner of autoscaling and health services, canaries, 15+ region builds, and on-call; cut region build time from 2 weeks to 3 days and saved \$200K by fixing an autoscaling bug. AI-first engineer using Claude Code and Amazon Q Developer/Kiro daily, cutting canary delivery from 4 weeks to 1 week.

WORK EXPERIENCE

Amazon Web Services (AWS) | Software Development Engineer

Dec 2024 – Present

Skills: Scala, Java, Python, TypeScript, C, Transit Gateway, VPC, VPN, DynamoDB, CloudFormation

Herndon, Virginia

- Develop and operate Autoscaler and HealthDecider, the Hyperplane control-plane services that scale Transit Gateway's data-plane fleet with traffic demand and detect unhealthy hosts for automated replacement.
- Saved **\$200K** in infrastructure costs by finding and fixing a latent Autoscaler bug that silently blocked cold scaling (scale-in of under-utilized capacity), restoring automatic right-sizing of the fleet.
- Built Transit Gateway in **15+ AWS Regions** and automated zonal (per-AZ) builds, cutting region build time from **2 weeks to 3 days** and bringing Transit Gateway to customers in new Regions sooner.
- Safely decommissioned Transit Gateway from Regions and AZs being phased out, with no operational disruption.
- Hardened the JIT migrator moving customers to a new proprietary data plane with zero packet loss: handled edge cases and added a circuit breaker that stops a broken metrics pipeline from triggering fleet-wide cold scaling.
- Built 4+ canaries that run multi-VPC topologies with VPC, VPN, and Direct Connect attachments to continuously test Transit Gateway features (appliance mode, long-running and always-changing stacks), catching regressions early.
- Cut canary delivery time from **4 weeks to 1 week** by building canaries with AI (Claude Code, Amazon Q Developer/Kiro) from detailed, context-rich prompts, then validating and testing them end to end.
- Resolved Transit Gateway issues in isolated, air-gapped AWS partitions with no direct access to hosts or logs, finding root cause by guiding on-site engineers who were new to the service through targeted diagnostic questions.
- Built operator tooling that replaces manual, unsupervised DynamoDB edits with deterministic, scoped commands, preventing accidental deletes or changes in production.
- Serve as on-call engineer for a critical networking service, triaging and resolving operational issues end to end.

IU School of Education | Research Assistant

Sep 2023 – Aug 2024

Skills: Python, NumPy, scikit-learn, Whisper, pyannote, Unity 3D, C#, C++

Bloomington, Indiana

- Built an AR app (Unity 3D, ARCore, ARKit) used by 2,000+ students in 3 schools, raising STEM engagement by 30%.
- Developed an ML model (NumPy, scikit-learn) for predictive analytics on 6,571 features (MSE 19.05) and a Whisper + pyannote speech-to-text pipeline with speaker diarization and sensitive-data masking.

Harman International (Samsung) | Software Engineer, Build & Automation

Sep 2020 – Jul 2022

Skills: Jenkins, JFrog Artifactory, SonarQube, Docker, Kubernetes, Terraform, Prometheus, Grafana, Python

Bangalore, India

- Built and maintained Jenkins CI/CD pipelines and jobs (freestyle, pipeline, multibranch, parameterized) for Audi and Porsche next-gen infotainment software, with JFrog Artifactory for build artifacts, reducing deployment time by 30%.
- Automated code-quality gates and tests with SonarQube, Python, and Bash, reducing production defects by 25%.
- Monitored build and service health on Prometheus/Grafana dashboards and managed AWS infrastructure with Terraform and Kubernetes, reducing downtime by 35% and cloud spend by 15%.

TECHNICAL SKILLS

Languages:	Scala, Java, Python, TypeScript, C, C++, Bash, SQL, JavaScript, C#, Ruby
Networking:	Transit Gateway (attachments, route tables), VPC, Site-to-Site VPN, Direct Connect, TCP/IP, BGP
AWS:	EC2, Lambda, DynamoDB, CloudWatch, CloudFormation, S3, SNS, Athena, Service Quotas, IAM, EKS, ECS
DevOps:	Jenkins, Git, Docker, Kubernetes, Terraform, Ansible, SonarQube, JFrog Artifactory, Prometheus, Grafana
AI Tools:	Claude Code, Amazon Q Developer, Kiro, ChatGPT, GitHub Copilot, prompt engineering, agentic coding
Technologies:	Spring Boot, Flask, Django, React, Node.js, .NET Core, MySQL, PostgreSQL, MongoDB, Redis

EDUCATION

Indiana University Bloomington

Aug 2022 – May 2024

Master of Science in Computer Science, **GPA: 3.7/4**

Bloomington, Indiana

Coursework: Computer Networks, Cloud Technologies, Enterprise Software Platforms, Machine Learning

JSS Science and Technology University

Jun 2016 – May 2020

Bachelor of Engineering in Computer Science, **CGPA: 8.9/10**

Mysore, India